Nuclei

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Nuclear Parameters

NUCLEUS

The nucleus is central part of atom having all the positive charge of atom in it and most of the mass of the atom.

Discoverer : Rutherford

Constituents : Neutrons(n) and Protons(p) [unanimously known as nucleons]

Representation : ${}_{Z}X^{A}$ or ${}^{A}_{Z}X$

Where X ⇒ Atom Symbol

Z ⇒ Atomic number which is equal to number of protons

A ⇒ Atomic mass number which is total number of nucleons

A = number of protons +neutrons

Atomic Mass Number :

It is defined as total number of nucleons. It is denoted by 'A'.
Let; Z = number of proton, N = number of neutron\
then

$\boxed{A = Z + N}$

Atomic Mass Unit (a.m.u.) :-

1 amu = $\frac{1}{12^{th}}$ of mass the of ${}_{6}C^{12}$ atom.

In ${}_{6}C^{12}$ atom

No of proton = 6

No of neutron = 6

Mass of proton (m_p) = 1.672×10^{-27} kg.

Mass of neutron (m_r) = 1.674×10^{-27} kg.

Since ; $m_p \approx m_n$

So: 1 amu = $\frac{1}{12} \times (6m_p + 6m_n)$

$= \frac{1}{12}(6m_p + 6m_p)$

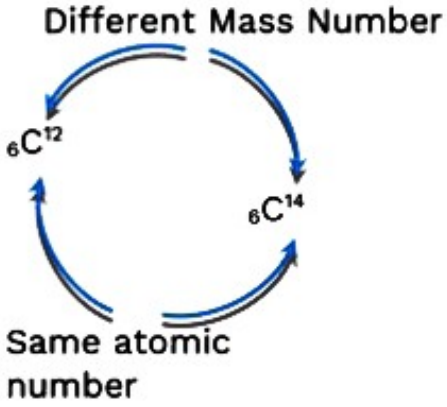
$= \frac{1}{12} \times 12\ m_p$

Conclusion :-
1 a.m.u. = mass of 1 proton
= mass of 1 neutron

SOME DEFINITIONS:

Iso topes : Atoms have same atomic number with different atomic mass number

Ecample :



Isobars : Atoms have same atomic mass number but different atomic number.

$_6C^{14}$; $_7N^{14}$

Isotones : Atoms have some number of neutron.

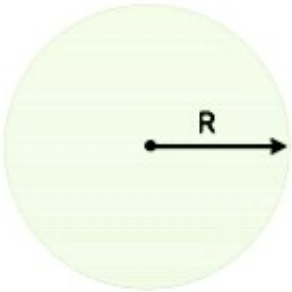
$_6O^{14}$; $_8O^{16}$

No of neutron = 14 – 6 = 16 – 8 = 8.

SIZE OF NUCLEI

Size of Nucleus :-

The size of nucleus is in the order of 10^{-15} m.



Nucleus

Radii of nucleus (R) = $R_0 (A)^{1/3}$

Where ;

$R_0 = 1.1 \times 10^{-15}$ m

A = Atomic mass of atom.

The credit of discovery of nucleus goes to Rutherford.

Density of Nucleus :-

$$\text{Density} = \frac{\text{mass of nucleus}}{\text{volume of nucleus}}$$

$$= \frac{A \times m_p}{\frac{4}{3}\pi R^3}$$

$$= \frac{A \times 1.672 \times 10^{-27}}{\frac{4}{3}\pi \times \left(R_0 \times A^{\frac{1}{3}}\right)^3}$$

$$= \frac{1.672 \times 10^{-27}}{\frac{4}{3}\pi \times R_0^3} \approx 2 \times 10^{17}\,\text{kg}$$

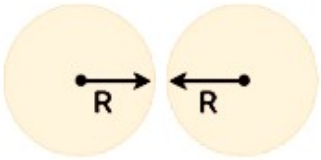
From above calculation we observed that nucleus has extremely high density.
Note : Since nuclear density is independent of A and Z. So almost all atoms have same nuclear density.

Q. **Find electrostatic potential energy between two 6C12 nuclei when they touch each other on nucleus surface.**

Sol. $R = R_0(A)^{1/3}$

$R = 1.1 \times 10^{-15} \times (12)^{1/3}$

$R = 2.75 \times 10^{-15}$ m.



Electrostatic Potential Energy

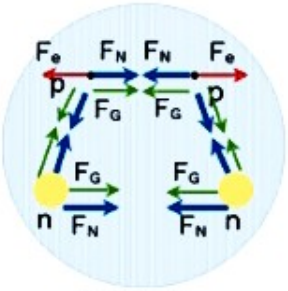
$U = \frac{1}{4\pi \epsilon_0} \times \frac{q_1 q_2}{r^2}$ q_1 = 6 × charge on one proton.

q_2 = 6 × charge on one proton.

$$U = 9 \times 10^9 \times \frac{\left(6 \times 1.6 \times 10^{-19}\right)^2}{\left(2 \times 2.75 \times 10^{-15}\right)^2}$$

$$\boxed{U = 1.5 \times 10^{-12}\,\text{J}}$$

NUCLEAR FORCE AND STABILITY OF NUCLEUS



Nucleus

F_e = Electrostatics Force.

F_G = Gravitational Force.

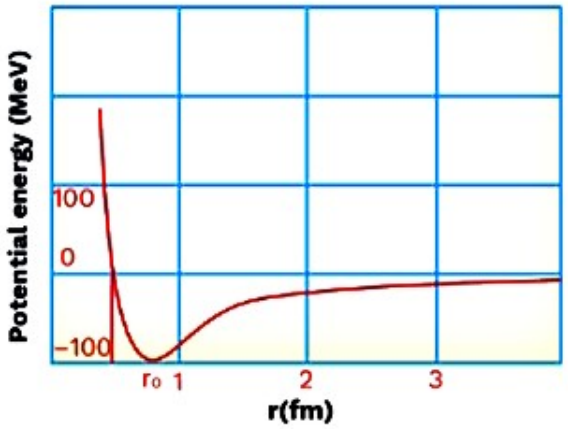
F_N= Nuclear force

Nucleons are bounded together inside the nucleus. Since nucleus is very very small so there is a very strong electrostatic repulsion acting between protons and protons. But still many atoms have a stable nucleus. Clearly there is some attractive force present between the nucleons which holds the nucleons together inside such a small space so that the nucleus of the atom is stable.

This force must be strong enough to overcome the repulsive Coulomb force between the (positively charged) protons and to bind both neutrons and protons into the tiny nuclear volume. Experiments suggest that this strong force, as it is simply called, has the same character between any pair of nuclear constituents, be they neutrons or protons.

The "strong force" has a short range, roughly equal to 10–15 m. This means that the attractive force between pairs of nucleons drops rapidly to zero for nucleon eparations greater than a certain critical value. Nuclear force is also called short range force.

The nuclear forces are very short-range forces and it can be inferred from a rough plot of the potential energy between two nucleons as a function of distance as shown in the figure below.

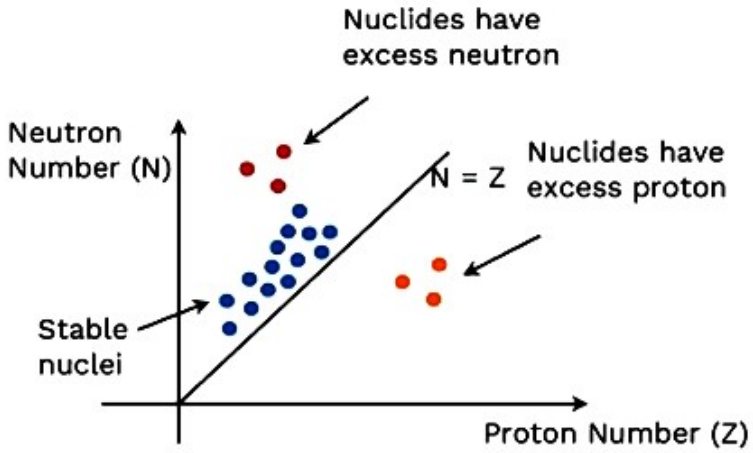


For a separation greater than r0, the force is attractive and for separations less than r0, the force is strongly repulsive.

Nuclear Stability Curve

Figure shows a curve between no of proton (Z) and no of neutron (N).

$\frac{N}{Z}$ = 1.6 for Heaviest stable nuclides



The nuclides which have excess number of proton or neutron are unstable and decay with time. According to the laws of radioactive disintegration. These nuclides are called radioactive nuclide.

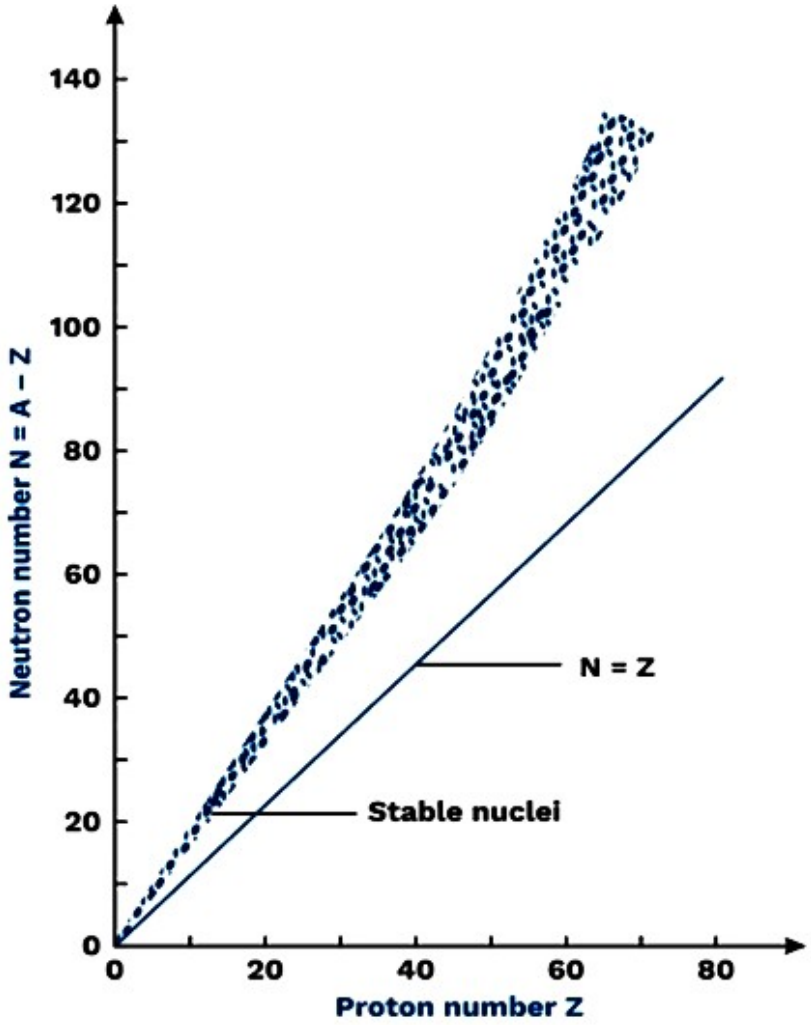
A graph showing the number of protons(Z) and the number of neutrons (N) is as shown below. A solid line then represents the equation N = Z. For all the known nuclei to mankind, the graph between N and Z is plotted and we get the dotted data as shown in the figure.

We know that protons repel reach other inside a nucleus. This repulsive force is a long range force and it can be observed in all the protons inside the nucleus. But the counter balancing strong nuclear force only attracts the neighbouring protons or neutrons. So, to keep the balance on the repulsive force, the number of neutrons are needed to be more than the number of protons in the nucleus.

A large number of nuclei fall under this category where the number of neutrons is more than the number of protons as shown in the graph.

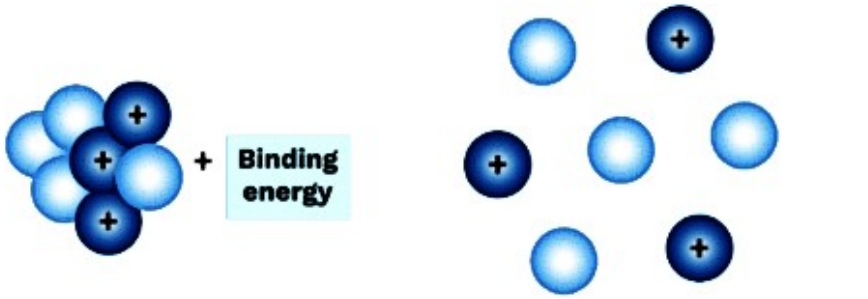
On moving towards the heavier nuclei, there comes a point where the balance of attractive and repulsive forces cannot be maintained as the strong forces have short range. So, there is a limitation to the number of neutrons inside the nucleus.

Bismuth $^{209}_{83}Bi$ is the heaviest naturally stable nucleus. All nuclei above Bi are unstable and readily disintegrate and rearrange their internal structure. This phenomenon is called radioactivity, which was discovered by Henri Becquerel in 1896.



NUCLEAR BINDING ENERGY

Energy required to separate nucleus or energy released at time of nucleus formation more stable nucleus more energy required to break apart. This required energy is called Binding energy. The mass of each element atom is less than the sum of masses of its constituent particles.



Nucleus (smaller mass) **Spherical nucleons (greater mass)**

Δm = Mass defect

= Mass of nucleons – Mass of nucleus

$= (m_n + m_p + m_e) - m_{nu}$

Since : $m_e << << << m_p$

So; $\boxed{(\Delta m) = (m_p + m_n) - m_{nu}}$

MASS-ENERGY EQUIVALENCE

$\boxed{E = (\Delta m)C^2}$

1 amu = 1.656×10^{-27} kg

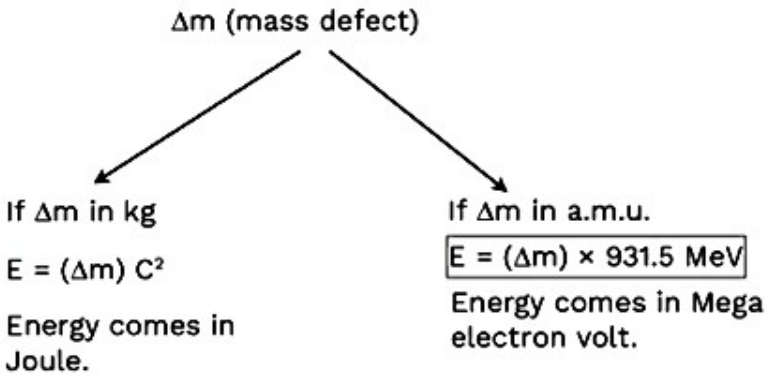
If in a reaction 1 amu mass is converted into energy, then using Einstein's mass-energy relationship the amount of energy released is

$$\Delta E = \Delta mc^2 = (1.656 \times 10^{-27})(3 \times 10^8)^2 J$$

$$= \frac{(1.656 \times 10^{-27})(3 \times 10^8)^2}{1.6 \times 10^{-19}} eV$$

$$= 931.5 \times 10^6 eV = 931.5 \text{ MeV}$$

Thus, we can say that 1 amu mass is equivalent to 931.5 MeV energy.



Q. **Calculate the binding energy He-Nucleus.**
Given that
Mass of Helium Nucleus = 4.001055 a.m.u.
Mass of Proton = 1.007276 a.m.u.
Mass of Neutron = 1.008668 a.m.u

Sol. Mass of nucleon = $m_n + m_p$

$= 2 \times (1.008668) + 2 \times (1.007276)$

$= 4.031888$

Mass of He-Nucleus = 4.001055

Mass of defect = $m_{nucleon} - m_{nucleus}$

$= (4.031888) - (4.001055)$

$= 0.030833$

$E = (\Delta m) \times 931$ MeV

$= (0.030833) \times 931.5$ MeV

Q. **The atomic mass of ^4_2He is 4.0026 u and the atomic mass of ^1_1H is 1.0078 u.**
Using atomic mass units instead of kilograms, obtain the binding energy of the ^4_2He nucleus. (Given mass of neutron = 1.0087 u)

Sol. Nuclear reaction

$2m_p + 2m_n \rightarrow m_{He}$ + Binding Energy

Mass of proton, m_p = (atomic mass of hydrogen) – (mass of electron) = $1.0078 - m_e$

Mass of He nucleus, m_{He} = (Atomic mass of ^4_2He) – (mass of two electrons) = $4.0026 - 2m_e$

Mass defect, $\Delta m = 2m_p + 2m_n - m_{He} = 2(1.0078 - m_e) + 2(1.0087) - (4.0026 - 2m_e)$

$\Delta m = 2(1.0078) + 2(1.0087) - 4.0026 = 0.0304$ u

Binding energy = $0.0304 \times 931.5 = 28.32$ MeV

BINDING ENERGY PER NUCLEON

$$\text{Binding Energy Per Nucleon} = \frac{\text{Binding Energy of Nucleus}}{\text{Number of Nucleon}}$$

$$\boxed{(BE)_N = \frac{\Delta E}{N}}$$

This value $(BE)_N$ gives the criterion of stability among different elements. Higher the binding energy per nucleon, higher is the stability of the nucleus.

Q.

A	B
B.E = 28 MeV	**BE = 24 MeV**
Number of nucleon = A = 7	**Number of Nucleon (A) = 4**
$(B \cdot E)_A = \frac{28}{7}$ MeV	$(BE)_B = \frac{24}{4}$
= 4 MeV	**= 6 MeV**

Sol. From above calculated data B is more stable than A although A has higher Binding Energy.

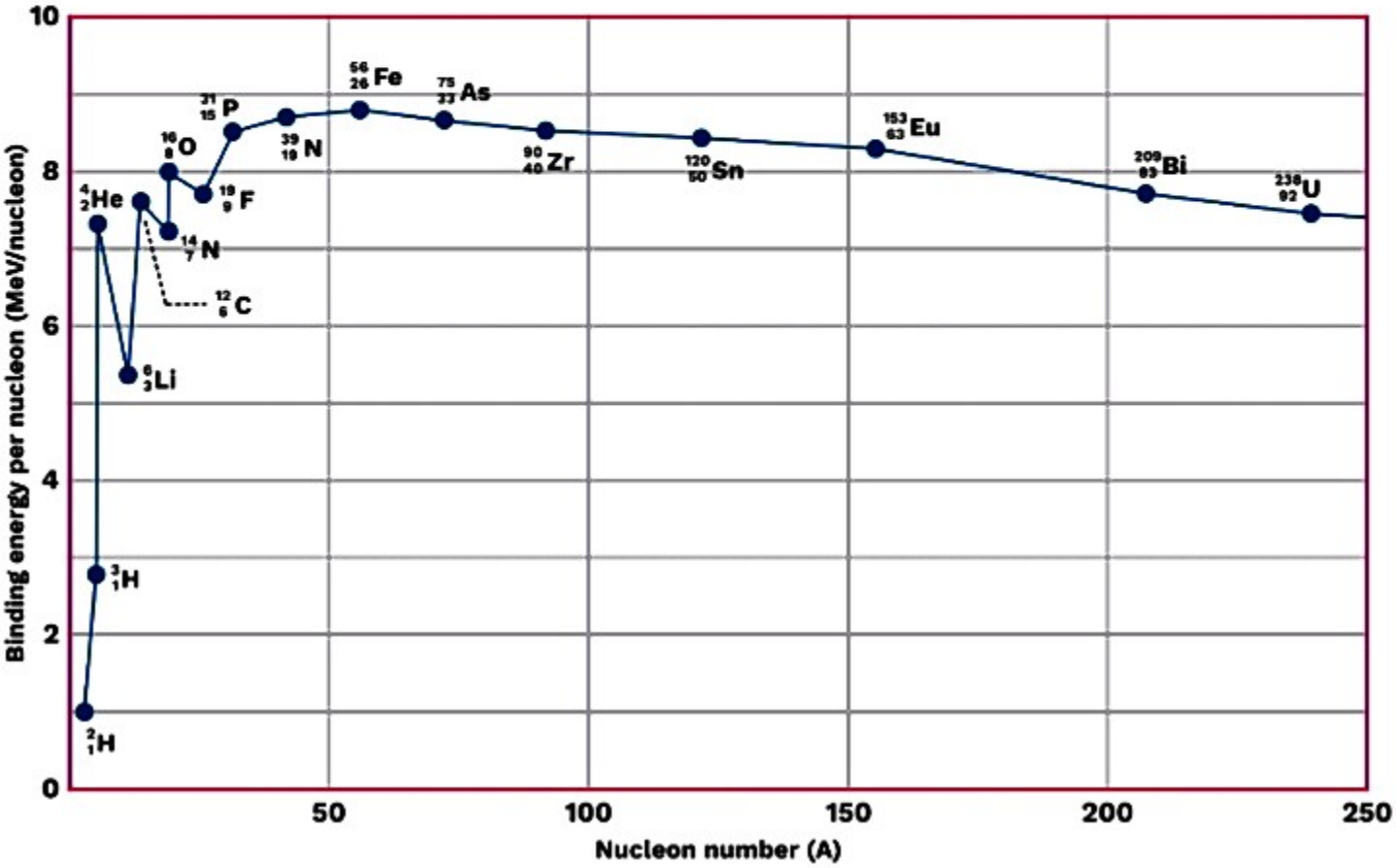
Q. **Following data tells about 3 nuclei P, Q and R. Arrange them in decreasing order of stability**

	P	Q	R
Atomic mass number (A)	**10**	**5**	**6**
Binding Energy (MeV)	**100**	**60**	**66**

Sol. $\left(\frac{B.E}{A}\right)_P = \frac{100}{10} = 10 \Rightarrow \left(\frac{BE}{A}\right)_Q = \frac{60}{5} = 12$

$\left(\frac{B.E}{A}\right)_R = \frac{66}{6} = 11$ $\therefore$ Stability order is Q > R > P.

Binding Energy Variation per Nucleon V/s Mass Number



A graph showing the binding energy per nucleon with the nucleon number is plotted as shown. It is found that the binding energies per nucleon for 4He, 12C, and 16O are relatively higher as compared to 6Li, 14N and 19F respectively. Higher the binding energy per nucleon, more is the stability of the nucleus.

This difference is due to presence of nuclear energy levels inside the nucleus. Each nuclear energy level can have two neutrons of opposite spin or two protons of opposite spin.

Energy levels in nuclei are filled in sequence similar to the energy levels in atoms in order to achieve maximum stability or minimum energy.

Also, the configurations of opposite spins in nucleons with pairs in nuclear energy levels are more stable. This explains the reason of more stability of 4He, 12C, and 16O compared to their neighbouring elements.

26Fe56 has maximum binding energy per nucleon of 8.8 MeV.

For medium nuclei, binding energy per nuclei is medium and thus they have high stability.

Fission :- The heavier nuclei split into medium nuclei being unstable. It is called fission process.

Fusion :- The lighter nuclei have tendency to fuse into a medium nucleus being unstable this process is called fusion.

Q. **Find the binding energy of the nuclei $^{56}_{26}Fe$ and $^{209}_{83}Bi$ in units of MeV from the following data:**

$m_p = 1.007825\,u$, $m_n = 1.008665\,u$

$m(^{56}_{26}Fe) = 55.934939\,u$, $m(^{209}_{83}Bi) = 208.980388\,u$

Which nucleus from the above has greater binding energy per nucleon?

Sol. In $^{56}_{26}Fe$ nucleus, there are 26 protons and (56 – 26) = 30 neutrons.

Mass defect $= 26 \times m_p + 30 m_n - m(^{56}_{26}Fe)$

$= 26(1.007825\,u) + 30(1.008665\,u) - 55.934939\,u$

$= 0.528461$

Binding energy $= 0.528461 \times 931.5$ MeV $= 492.26$ MeV (as 1 u = 931.5 MeV)

Binding energy per nucleon $= \dfrac{492.26 \text{ MeV}}{56} = 8.79$ MeV

In $^{209}_{83}Bi$ nucleus, there are 83 protons and (209 – 83) = 126 neutrons.

Mass defect, $\Delta m = 83 \times m_p + 126 \times m_n - m(^{209}_{83}Bi)$

$= 83(1.007825\ u) + 126(1.008665\ u) - 208.980388\ u$

$= 1.760872\ u$

Binding energy $= 1.760872 \times 931.5$ MeV $= 1640.3$ MeV

Binding energy per nucleon $= \dfrac{1640.3}{209} = 7.85$ MeV

Thus, $^{54}_{26}Fe$ has greater binding energy per nucleon than $^{209}_{83}Bi$.

Q. **The binding energies per nucleon for deuteron ($_1H^2$) and helium ($_2He^4$) are 1.1 MeV and 7.0 MeV, respectively. Calculate the energy released when two deuterons fuse to form a helium nucleus ($_2He^4$).**

Sol. The equation of fusion is represented as

$_1H^2 + {}_1H^2 \rightarrow {}_2He^4$ + Energy

The binding energy per nucleon of deuteron is 1.1 MeV

Therefore, the net initial binding energy = 4 × 1.1 = 4.4 MeV

The binding energy per nucleon of $_2He^4$ is 7.0 MeV

Therefore, the net final binding energy is 4 × 7.0 = 28.0 MeV

Hence, energy released = 28.0 MeV – 4.4 MeV = 23.6 MeV

Radioactivity

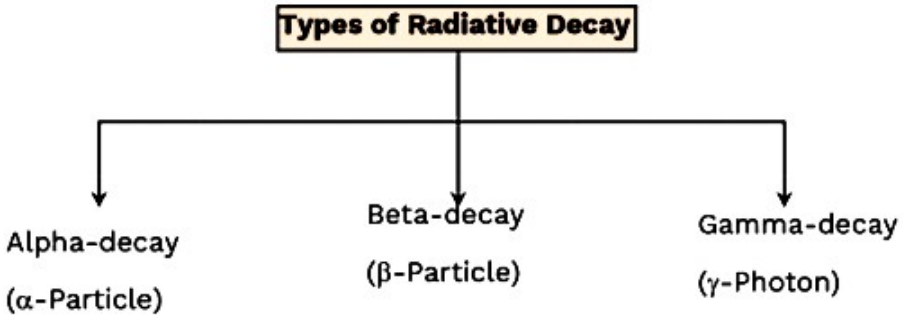
The nucleus of certain atoms are unstable in nature. This is due to improper N/Z ratio. Inside a nucleus, electrostatic attraction is counterbalanced by short-range strong nuclear forces and the nucleus becomes stable. Despite the forces being balanced, many nuclides are unstable because of nuclear size or the limited range of nuclear forces Nucleus gain stability by emitting energy and sub atomic particle. An spontaneous disintegration on decay of nucleus into more stable nucleus called radioactivity. It was discovered by Henry Becquerel.

Important Point :-

(1) There is no fixed time at which nucleus must decay.

(2) Radio-active is a property of several naturally occurring element as well as artificially produced isotopes of the element.

(3) Discovered by Henry Becqurel.



When an radioactive disintegration spontaneously, certain kinds & particles or high energy photons are released. These particles are collectively called 'rays'.

Alpha particle (α):

Nuclei with mass number more than 210 undergo α-decay.

In this decay, mass number decreases by four and atomic number by two along with γ-Radiation.

Daughter Nucleus goes in upper excited state.

$${}_{Z}X^{A} \xrightarrow{\alpha\text{-decay}} {}_{Z-2}{}^{*}Y^{A-4} + {}_{2}He^{4} \longrightarrow {}_{Z-2}Y^{A-4} + {}_{2}He^{4} + (\gamma - \text{Photon})$$

Parent Nucleus

α – particles (Helium Nucleus)

All energy (Q value) that we get due to mass defect will be in form of kinetic energy of daughter nuclei and gamma radiation.

$$\boxed{Q \text{ Value} = (M_X - M_Y - M_{He})\, c^2 \text{Joule.}}$$

$$\boxed{Q \text{ Value} = (M_X - M_Y - M_{He}) \times 9315 \text{ MeV}}$$

This energy is available in the form of kinetic energy of the products.

Kinetic energy of α-particle $= \dfrac{(A-4)Q}{A}$

Kinetic Energy of Y $= \dfrac{4Q}{A}$.

Q. **Find the Q value and the kinetic energy of the emitted α-particle in the α-decay of $_{88}Ra^{226}$.**
Given : Mass of $_{88}Ra^{226}$ = 226.02540 a.m.u.
Mass of $_{88}Rn^{222}$ = 222.01750 a.m.u.

Sol. $_{88}Ra^{226} \xrightarrow{\alpha\text{-decay}} {}_{86}Rn^{222} + {}_{2}He^{4} + Q$

$Q = [m_{Ra} - m_{Rn} - m_{He}] \times 931.5$ MeV

$= [226.02540 - 222.01750 - 4.00260] \times 931.5$

$Q = 4.49$ MeV

Kinetic energy of α - particle $= \dfrac{(A-4)}{A} \times Q$

$= \dfrac{(226-4)}{226} \times 4.94\,\text{MeV}$

$KE_{He} = 4.85\,\text{Mev}$.

Q. **Find the Q-value and the kinetic energy of the emitted α-particle in the α-decay of (a) $^{226}_{88}Ra$ and (b) $^{220}_{86}Rn$.**

Given: $m(^{226}_{88}Ra) = 226.02540$ u

$m(^{222}_{86}Rn) = 222.01750$ u

$m(^{216}_{84}Po) = 216.00189$ u

Sol. (a) As $^{226}_{88}Ra \rightarrow {}^{222}_{86}Rn + {}^{4}_{2}He + Q$

$Q = [m(^{226}_{88}Rn) - m(^{222}_{86}Rn) - m(^{4}_{2}He)]$

$= [226.02540 - 222.01750 - 4.00260]u = 0.0053$ u

$= 0.0053 \times 931.5$ MeV $= 4.94$ MeV

KE of the α-particle

$$K_\alpha = \frac{(A-4)}{A} \times Q = \frac{(226-4)}{226} \times 4.94 \text{ MeV} = 4.84 \text{ MeV}$$

(b) Following same method as in (a),

$$Q = 0.00688 \times 931.5 \text{ MeV} = 6.41 \text{ MeV}$$

KE of the α-particle,

$$K_\alpha = \frac{(A-4)}{A} Q = \left(\frac{220-4}{220}\right) \times 6.41 \text{ MeV} = 6.29 \text{ MeV}$$

β–decay

- β^- - decay
- β^+ - decay

β^--decay : The basic nuclear process underlying β^- - decay is the conversion of neutron to proton.

$$n \longrightarrow P^+ + e^- + \bar{\nu}$$

Deacy equation :-

Nucleus is in excited state

$${}_{z}X^{A} \longrightarrow {}_{z+1}{}^{*}Y^{A} + {}_{-1}\beta^0 + \bar{\nu} \longrightarrow {}_{z+1}Y^{A} + {}_{-1}\beta^0 + \bar{\nu} + \gamma \text{ Photon}$$

Important Point :-

(1) After β^--decay, N/P Ratio decreases.

(2) ${}_{-1}\beta^0$ always comes out from the nucleus along with anti-neutrino $(\bar{\nu})$.

Q. Consider the beta decay $^{198}Au \rightarrow {}^{198}Hg^* + \beta^- + \bar{\nu}$ where $^{198}Hg^*$ represents a mercury nucleus in an excited state at energy 1.088 MeV above the ground state. What can be the maximum kinetic energy of the electron emitted? The atomic mass ^{198}Au 197.968233 u and that of ^{198}Hg is 197.966760 u.

Sol. If the product nucleus ^{198}Hg is formed in its ground state, the kinetic energy available to the electron and the antineutrino is $Q = [m(^{198}Au) - m(^{198}Hg)]\,c^2$
As $^{198}Hg^*$ has energy 1.088 MeV more than ^{198}Hg in ground state, the kinetic energy actually available is $= [m(^{198}Au)] - m(^{198}Hg)]\,c^2 - 1.088$ MeV

$$= (197.968233\,u - 197.966760\,u)\left(931.5\,\frac{MeV}{u}\right) - 1.088\ MeV$$

$= 1.3721$ MeV $- 1.088$ MeV $= 0.2841$ MeV.

This is also the maximum possible kinetic energy of the electron emitted.

β^+ - decay : The basic nuclear process underlying β^+-decay is the conversion of proton into neutron.

$$_1P^1 \longrightarrow {}_0n^1 + {}_{+1}\beta^0 + \nu$$

Proton neutron Positron neutrino

Deacy equation :-

$$^A_ZX \longrightarrow {}^A_{Z-1}Y^* + {}_{+1}\beta^0 + \nu \longrightarrow {}^A_{Z-1}Y + {}_{+1}\beta^0 + \nu + \gamma\,\text{photon}.$$

Important Point :-

(1) After $_{+1}\beta^0$ decay, N/P Ratio increases.

(2) $_{+1}\beta^0$ comes out from the nucleus with neutrino.

Note:-

In proton to neutron **conversion** as mass increases**, it then requires** energy for β^+ decay to take place.

β **+** decay is being **a rare** process which takes place in the **nucleus,** where a proton **draws** energy from the nucleus itself.

(3) $_{-1}\beta^0$ can written as $_{-1}e^0$ (electron)

(4) $_{+1}\beta^0$ can written as $_{+1}e^0$ (positron)

(5) Electron and positron are antiparticles.

Q. **Calculate the Q-value in the following decays:**

(a) $^{19}O \rightarrow {}^{19}F + e^- + \bar{\nu}$

(b) $^{25}Al \rightarrow {}^{25}Mg + e^+ + \bar{\nu}$

The atomic masses needed are as follows:

^{19}O	^{19}F	^{25}Al	^{25}Mg
19.003576 u	18.998403 u	24.990432 u	24.985839 u

Sol. (a) The Q-value of β^- - decay is

$$Q = [m(^{19}O) - m(^{19}F)]\,c^2$$

$$= [19.003576\ u - 18.998403\ u]\ (931.5\ MeV/u) = 4.819\ MeV$$

(b) The Q-value of β^+ -decay is

$$Q = [m(^{25}Al) - m(^{25}Mg) - 2m_e]\,c^2$$

$$= \left[24.99032\,u - 24.985839\,u - 2\times 0.511\frac{MeV}{c^2}\right]c^2$$

$$= (0.004593\ u)\ (931.5\ MeV/u) - 1.022\ MeV$$

$$= 4.278\ MeV - 1.022\ MeV = 3.256\ MeV.$$

γ-decay :- Similar to an atom nucleus also have certain energy levels and nucleus occupy them. After α-decay or β-decay, daughter nucleus may be in excited state and return to ground state by emitting photon of high energy (MeV order) called photon.

γ-decay :-

$$_zX + {}^{+A} \longrightarrow {}_zX^A + \nu - \text{photon}$$

* Shows excited state.

$$^{12}_5B = {}^{12}_6C + e^- + \bar{\nu}$$

$$^{11}_6C = {}^{12}_6C + \gamma$$

Important Point :-

(a) γ-emission do not change the structure of nucleus

(b) No change in Z or A.

Note:

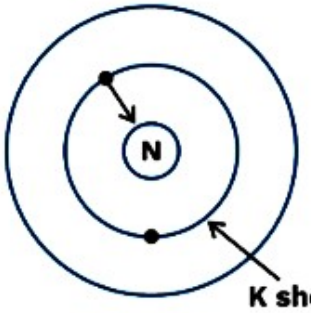
- Ionisation power $\Rightarrow \alpha > \beta > \gamma$.
- Penetrating power $\Rightarrow \gamma > \beta > \alpha$.

Q. **Find the wavelength of the 0.186 MeV γ-ray photon emitted by radium $^{226}_{88}Ra$?**

Sol. The wavelength of the photon is

$$\lambda = \frac{hc}{\Delta E} = \frac{12400\,eV - A^\circ}{0.186\times 10^6\,eV} = 6.67\times 10^{-12}m$$

The K–Electron Capture :-



In K-Electron capture a nucleus capture one of the inner orbital electron and a proton transform into a neutron.

$$_zX^A + -_{-1}e^0 \left(K - capture\right) \rightarrow _{z-1}X^A + \nu + X - Ray$$

Hence K-Capture is like a positron decay in both N/P ratio increases. In this event vacancy is created in R-shell to fill up the vacancy, electron transition takes place and X-Rays are emitted.

Properties of antineutrino ($\bar{\nu}$) and neutrino (ν) :

They have rest mass equal to zero or, at most, the mass equivalent of a few electronvolts.
Speed = c (or nearly equal to c)
Energy, $E = mc^2$
They are chargeless (neutral).
They have spin quantum number, $s = \pm \frac{1}{2}$

Considering the emission of antineutrino, the equation of β^- - decay can be written as

$$_zX^A \longrightarrow _{z-1}Y^A + _{-1}e^0 + Q + \bar{\nu}$$

They are not electromagnetic in nature as is the photon, the neutrino can pass unimpeded through vast amounts of matter.
Production of antineutrino along with the electron helps to explain the continuous spectrum because the energy is distributed randomly between electron and $\bar{\nu}$ and it also helps to explain the spin quantum number balance (p, n and ± e each has spin quantum number ± 1/2).
During β^- - decay, inside the nucleus a neutron is converted to a proton with emission of an electron and antineutrino.

$$n \rightarrow p + _{-1}e^0 + \bar{\nu}$$

Let, M_x mass of atom $_zX^A$

M_y = mass of atom $_{z+1}Y^A$

m_e = mass of electron

Q value = $[(M_X - Zm_e) - \{(M_Y - (z + 1)\, m_e) + m_e\}]\, c^2 = [M_x - M_Y]\, c^2$

Radioactive decay law :-

(Given by Rutherford and Soddy)

This law relates activity to the no of parent nuclei at an instant.

It is states as , "At any instant activity of a radio-active substance is directly proportional to the number of undecayed nuclei in the substance at that instant."

Let at t = 0; there were N_0 nuclei & at time 't' ; N are left undecayed from time t to t + dt. If dN nuclei will more decay :→

Activity of substances at time t is

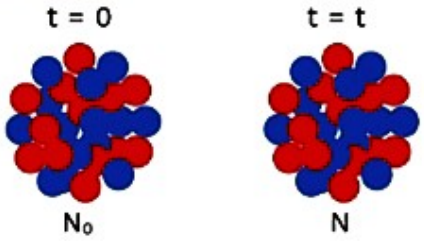
$$A_c = -\frac{dN}{dt}$$

According to radio-active decay law

$$-\frac{dN}{dt} \propto N$$

$$\boxed{-\frac{dN}{dt} = \lambda N}$$

λ = Proportionality decay constant.



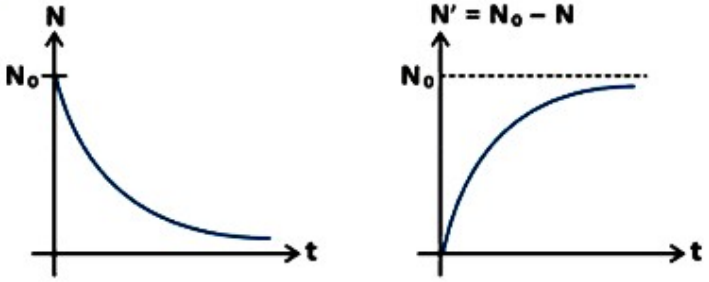
$$\int_{N_0}^{N} \frac{dN}{N} = -\int_0^t \lambda dt \qquad \left|\ell n(N)\right|_{N_0}^{N} = -\lambda \times \left|t\right|_0^t$$

$$\ell n_e\left(\frac{N}{N_0}\right) = -\lambda(t-0)$$

$$\ell n_e\left(\frac{N}{N_0}\right) = -\lambda t$$

$$e^{-\lambda t} = \frac{N}{N_0}$$

Where λ is the decay constant for the radioactive materials.



Similarly :

(1) Mass of Radio-active element :-

(2) Activity is given by

Decay-Constant of a Radio-Active Element:-

$\Rightarrow$ At on instant activity (dps) of a Radio-Active is given as:-

$$A_c = \left|-\frac{dN}{dt}\right| = \lambda N \qquad \text{Where } \lambda = \text{decay constant}$$

N = Undecayed Nuclei-Present at that instant.

$$\text{Now ; } \lambda = \frac{\left|\left(\frac{dN}{dt}\right)\right|}{N} = \frac{\text{Activity of Sample}}{\text{Number of active Nuclei in Sample}}$$

λ = Activity per nuclei of a radio-active substance
It is constant
It is a measure of stability of a radio-active element
Hight value of λ is lens stable element.
λ high $\Rightarrow$ High activity
λ Low $\Rightarrow$ It is more stable element and activity is low.

Half life ($T_{1/2}$) :

It is defined as the time in which number of active nuclei becomes half.

$$N = N_0 e^{-\lambda t}$$

After one half life, $N = \frac{N_0}{2}$

$$\frac{N_0}{2} = N_0 e^{-\lambda t} \Rightarrow t = \frac{\ln 2}{\lambda} \Rightarrow \frac{0.693}{\lambda} = t_{1/2}$$

$$t_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

$$\boxed{T = \frac{\ell n(2)}{\lambda} = \frac{0.693}{\lambda}}$$

Alternative forms of radio-active decay equation:-

If a radio-active element X decays to Y with decay constant 'λ'.

$X \xrightarrow{\lambda} Y$ Decay equation is written as

$$N = N_0 \cdot e^{-\lambda t}$$

$$\frac{N}{N_0} = e^{-\lambda t}$$

Take natural log both side

$$\ell n_e\left(\frac{N}{N_0}\right) = \ell n_e\left(e^{-\lambda t}\right)$$

$$\ell n_e\left(\frac{N}{N_0}\right) = -\lambda t$$

From relation ; $\lambda = \frac{\ell n(2)}{t_{1/2}}$

$$\ell n_e\left(\frac{N}{N_0}\right) = -\frac{\ell n(2)}{t_{1/2}} \times (t)$$

$$\ell n_e\left(\frac{N}{N_0}\right) = -\left(\frac{t}{t_{1/2}}\right) \ell n_e(2)$$

$$\ell n_e\left(\frac{N}{N_0}\right) = \ell n_e(2)^{-\frac{t}{t_{1/2}}}$$

$$\frac{N}{N_0} = (2)^{-\frac{t}{t_{1/2}}}$$

$$\boxed{N = N_0\left(\frac{1}{2}\right)^{\frac{t}{t^{1/2}}}}$$

Let; n = number of half life $= \frac{t}{t_{1/2}}$

$$\boxed{N = N_0\left(\frac{1}{2}\right)^n}$$

$$N_0 \xrightarrow[\text{half life}]{\text{after 1st}} \frac{N_0}{2} \xrightarrow{2} N_0\left(\frac{1}{2}\right)^2 \xrightarrow{3} N_0\left(\frac{1}{2}\right)^3 \cdots\cdots \xrightarrow{n} N_0\left(\frac{1}{2}\right)^n$$

Similarly :-

$$\boxed{m = m_0\left(\frac{1}{2}\right)^n}$$

$$\boxed{A_c = A_0\left(\frac{1}{2}\right)^n}$$

Where; m = Instantaneous mass of sample
m_0 = Initial mass of sample
A = Instantaneous activity of sample
A_c = Initial activity of sample

Activity :

Activity is defined as the rate of radioactive decay of the nucleus.
It is signified by A or R, $A = \lambda N$
If a radioactive substance changes only by decay, then

$$A = -\frac{dN}{dt}$$

As in that case, $N = N_0 e^{-\lambda t}$

$$A = \lambda N = \lambda N_0 e^{-\lambda t}$$

$$A = A_0 e^{-\lambda t}$$

SI unit of activity : becquerel (Bq) which is basically same as 1 dps (disintegration per second)
Curie is the popular unit of activity described as :
1 curie = 3.7 × 1010 dps (It is also activity of 1 gm Radium)

Specific Activity:

Activity of unit mass of radioactive substance is called specific activity.

Average Life:

	A $\longrightarrow$	B
$t = 0$	N_0	0
$t = t$	$N_0 e^{-\lambda t}$	$N_0(1 - e^{-\lambda t})$

$$R = R_0 e^{-\lambda t}$$

$$N = N_0 e^{-\lambda t}$$

$$\frac{dN}{dt} = -\lambda N_0 e^{-\lambda t}$$

$$t_{av} = \frac{\int t(-dN)}{N_0}$$

$$= \frac{\int_0^\infty t(\lambda N_0 e^{-\lambda t})dt}{N_0}$$

$$\boxed{t_{av} = \frac{1}{\lambda}}$$

Q. **The half-life of ^{198}Au is 2.7 days. Calculate (a) the decay constant. (b) the average-life, and (c) the activity of 1.00 mg of ^{198}Au. Take atomic weight of ^{198}Au to be 198 g mol^{-1}.**

Sol. (a) The half-life and the decay constant are related as

$$T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

or
$$\lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{2.7 \text{ days}} = \frac{0.693}{2.7 \times 24 \times 3600 \text{s}}$$

$$= 2.9 \times 10^{-6} \text{ s}^{-1}$$

(b) The average-life is $t_{av} = \frac{1}{\lambda} = 3.9$ days.

(c) The activity is $A = \lambda N$. Now, 198 g of ^{198}Au has 6×10^{23} atoms. The number of atoms in 1.00 mg of ^{198}Au is

$$N = 6 \times 10^{23} \times \frac{1.0 \text{mg}}{198 \text{g}} = 3.03 \times 10^{18}$$

Thus, $A = \lambda N = (2.9 \times 10^{-6} \text{ s}^{-1})(3.03 \times 10^{18})$

$$= 8.8 \times 10^{12} \text{ disintegrations s}^{-1}$$

$$= \frac{8.8 \times 10^{12}}{3.7 \times 10^{10}} \text{Ci} = 240 \text{Ci}$$

Q. **A 50.0 g sample of carbon is taken from the pelvis bone of a skeleton and is found to have a ^{14}C decay rate of 200.0 decays min^{-1}. It is known that carbon from a living organism has a decay rate of decays/min^{-1} g^{-1} and that ^{14}C has a half-life of 5730 year = 3.01×10^9 min. Find the age of the skeleton.**

Sol. Let us start with the equation $N = N_0 e^{-\lambda t}$ and multiply both sides by λ to get $\lambda N = \lambda N_0 e^{-\lambda t}$ which is equivalent to

$$R = R_0 e^{-\lambda t} \quad \text{or} \quad \frac{R}{R_0} = e^{-\lambda t}$$

where R is the present activity and R_0 was the activity when the skeleton was a part of a living organism. We can solve for the time by taking the natural log of both sides of this equation.

$$\ln\left(\frac{R}{R_0}\right) = \ln(e^{-\lambda t}) = -\lambda t$$

$$t = -\frac{\ln\left(\frac{R}{R_0}\right)}{\lambda}$$

Because we are given the decay rate and mass of the sample, we can find R_0 as

$$R_0 = \left(15.0\,\frac{\text{decays}}{\text{min g}}\right)(50.0\text{g}) = 750\,\frac{\text{decays}}{\text{min}}$$

The decay constant is found as

$$\lambda = \frac{0.693}{T_{1/2}} = \frac{0.693}{3.01\times 10^{9}\,\text{min}} = 2.30\times 10^{-10}\,\text{min}^{-1}$$

Q. **At a given instant, there are 25% undecayed radioactive nuclei in a sample. After 10 s, the number of undecayed nuclei reduces to 12.5%. Calculate (a) mean life of the nuclei and (b) the time in which the number of decayed nuclei will further reduce to 6.25% of the reduced number.**

Sol. Half-life or radioactive sample, i.e., the time in which number of undecayed nuclei becomes half (T) is 10 s.

(a) Mean life, $\tau = \frac{T}{\log_e 2} = \frac{10}{0.693}\text{s} = 1.443\times 10 = 14.43\text{s}$

(b) The reduced number further reduces to 6.25% in n half-lives given by

$$\frac{N}{100} = \left(\frac{1}{2}\right)^n \Rightarrow \frac{6.25}{100} = \left(\frac{1}{2}\right)^n$$

$$\Rightarrow \quad \frac{1}{16} = \left(\frac{1}{2}\right)^n \Rightarrow \left(\frac{1}{2}\right)^4 = \left(\frac{1}{2}\right)^n \Rightarrow n = 4$$

Time, $t = 4T = 4\times 10 = 40$ s

Q. **The normal activity of living carbon-containing matter is found to be about 15 decays per minute for every gram of carbon. Suppose a specimen from Mohenjodaro gives an activity of 9 decays per minute per gram of carbon. Estimate the approximate age of the Indus-Vally civilization from the known half-life (=5730 years) of C^{14}.**

Sol. Let t be the age of the Indus-Valley civilization.

The rate of decay, i.e., R is given by

$$R \propto N \quad \text{or} \quad R = \lambda N$$

As $N = N_0 e^{-\lambda t}$, $R = \lambda N_0 e^{-\lambda t} = R_0 e^{-\lambda t}$

Thus, $$\frac{R}{R_0} = e^{-\lambda t} \qquad \ldots(i)$$

We are given that $$\frac{R}{R_0} = \frac{9}{15} = \frac{3}{5} \qquad \ldots(ii)$$

From Eqs. (i) and (ii), $e^{-\lambda x} = \frac{3}{5}$ or $e^{\lambda x} = \frac{5}{3}$

or $$\lambda t = \ln\left(\frac{5}{3}\right) \text{ or } t = \frac{\ln(5/3)}{\lambda}$$

or $$t = \frac{\ln(5/3)}{\ln 2} \times T_{1/2} = \frac{\log(5/3)}{\log 2} \times 5730\text{yr} \left(\text{as } T_{1/2} = \frac{\ln 2}{\lambda} \right)$$

or $$t = \frac{0.2219}{0.3010} \times 5730\text{yr} = 4224\text{yr}.$$

Q. **The half-life of ^{198}Au is given to be 2.7 days. Calculate**
(a) decay constant,
(b) mean lifetime and
(c) activity of 1.00 mg of ^{198}Au. Take the atomic mass 198Au to be 198 g/mol.

Sol. (a) The half-life and the decay constant are related as

$$t_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

or $$\lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{2.7\,\text{days}} = \frac{0.693}{2.7 \times 24 \times 3600\,\text{s}} = 2.9 \times 10^{-6}\,\text{s}^{-1}.$$

(b) The average-life is $t_{av} = \frac{1}{\lambda} = 3.9$ days.

(c) The activity is $A = \lambda N$. Now, 198 g of ^{198}Au has 6×10^{23} atoms.
The number of atoms in 1.00 mg of ^{198}Au is

$$N = 6 \times 10^{23} \times \frac{1.0\,\text{mg}}{198\,\text{g}} = 3.03 \times 10^{18}$$

Thus, $A = \lambda N = (2.9 \times 10^{-6}\ \text{s}^{-1})\ (3.03 \times 10^{18})$

$$A = 8.8 \times 10^{12} \text{ disintegrations/s} = \frac{8.8 \times 10^{12}}{3.7 \times 10^{10}}\,\text{Ci} = 240\,\text{Ci}.$$

Q. **Consider two radioactive substances initially having same number of nuclei, their mean lives are T_1 and T_2 respectively, such that $T_1 > T_2$. After how much time, the number of active nuclei in one substance will be $\frac{1}{e^2}$ times of number of active nuclei in other sample.**

Sol.

	P	Q	
	N_0	N_0	
Mean life	T_1	T_2	$T_1 > T_2$

$$N_P = N_0 e^{-\lambda_1 t} \qquad N_Q = N_0 e^{-\lambda_2 t}$$

$$N_Q = \frac{1}{e^2} N_P$$

$$N_0 e^{-\lambda_2 t} = \frac{1}{e^2} e^{-\lambda_1 t}$$

$$e^2 = e^{(\lambda_2 - \lambda_1)t}$$

$$2 = (\lambda_2 - \lambda_1)t \Rightarrow 2 = \left(\frac{1}{T_2} - \frac{1}{T_1}\right)t$$

$$t = \left(\frac{2}{\left(\frac{1}{T_2} - \frac{1}{T_1}\right)}\right) \Rightarrow t = \frac{2T_1T_2}{(T_1 - T_2)}$$

CARBON DATING

The radiocarbon technique utilizes the ${}^{14}_{6}C$ isotope of carbon, which undergoes β^- decay with a half-life of 5730 years. This isotope is present in the earth's atmosphere at an equilibrium concentration of about one atom for every 8.3×10^{11} atoms of normal carbon ${}^{12}_{6}C$. It is often assumed that this value has remained constant over the years because ${}^{14}_{6}C$ is created when cosmic rays interact with the earth's upper atmosphere, a production method that offsets the loss via β^- decay. Moreover, nearly all living organisms ingest the equilibrium concentration of ${}^{14}_{6}C$. However, once an organism dies, metabolism no longer sustains the input of ${}^{14}_{6}C$, and β^- decay causes half of the ${}^{14}_{6}C$ nuclei to disintegrate every 5730 years.

Q. **The normal activity of living matter containing carbon is about 15 decays per minute per gram of carbon. Assume that a specimen of MohenjoDaro gives an activity of 9 decays per minute per gram of carbon. Approximate age of the IndusValley civilization from the known half-life (= 5730 years) of C^{14}.**

Sol. Let t be the age of the Indus-Valley civilization.
The rate of decay, i.e., R is given by

$R \propto N$ or $R = \lambda N$

As $N = N_0 e^{-\lambda t}$, $R = \lambda N_0 e^{-\lambda t} = R_0 e^{-\lambda t}$

Where $R_0 = \lambda N_0$ is the initial rate of decay.

Thus, $$\frac{R}{R_0} = e^{-\lambda t} \quad ...(1)$$

We are given that $$\frac{R}{R_0} = \frac{9}{15} = \frac{3}{5} \quad ...(2)$$

From Eqs. (1) and (2), $e^{-\lambda t} = \frac{3}{5}$ or $e^{\lambda t} = \frac{5}{3}$

or $$\lambda t = \ln\left(\frac{5}{3}\right) \text{ or } t = \frac{\ln(5/3)}{\lambda}$$

or $$t = \frac{\ln(5/3)}{\ln 2} \times T_{1/2} = \frac{\log(5/3)}{\log 2} \times 5730\,\text{yr} \quad \left(\text{as } T_{1/2} = \frac{\ln 2}{\lambda}\right)$$

or $$t = \frac{0.2219}{0.3010} \times 5730\,\text{yr} = 4224\,\text{yr}$$

Q. **A radioactive nucleus undergoes a line of decay according to the following**

$$A \xrightarrow{\alpha} A_1 \xrightarrow{\beta^-} A_2 \xrightarrow{\alpha} A_3 \xrightarrow{\gamma} A_4$$

The mass number and atomic number of A are 180 and 72, respectively, what are these numbers for A_4?

Sol. The successive process may be expressed as

$$_{72}A^{180} \xrightarrow{\alpha} {}_{70}A_1^{176} \xrightarrow{\beta^-} {}_{71}A_2^{176} \xrightarrow{\alpha} {}_{69}A_3^{172} \xrightarrow{\gamma} {}_{69}A_4^{172}$$

Radioactive Equilibrium

In a radioactive series, if we talk about an intermediate element, it is produced due to the decay of its previous element and it decays to the next element of the series.

In a Radio-Active Series :-

$$\underset{t=t}{} \quad \underset{N_1}{A_1} \xrightarrow{\lambda_1} \underset{N_2}{A_2} \xrightarrow{\lambda_2} A_3 \xrightarrow{\lambda_3} .....$$

In a radio-active series, if we take about an intermediate element, it is produced due to the decay of its previous element and it decays to the next element.

Production rate of A_2 = Disintegration rate of A_1 = $\lambda_1 N_1$

$$A_1 \Rightarrow \frac{dN}{dt} \propto N_1$$

$$\text{Disintegration Rate}\left(-\frac{dN}{dt}\right) = \lambda_1 N_1$$

Now, Disintegration rate of A_2

$$A_2 \longrightarrow -\frac{dN}{dt} \propto N_2$$

$$\left(\frac{dN}{dt}\right) = \lambda_2 N_2$$

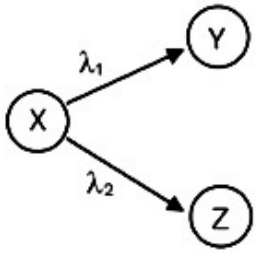
$$\lambda_1 N_1 \longrightarrow \boxed{A_2} \longrightarrow \lambda_2 N_2$$

So; If A_2 will be in ratio-active equilibrium then :-

$$\boxed{\lambda_1 N_1 = \lambda_2 N_2}$$

Rate of formation = Rate of disintegration

Simultaneous – decay mode of a radio-active element:-



If $\lambda_1 > \lambda_2$ probability of formation y is more than z,

If half-life of x is $T_{1/2}$ then;

$$\lambda_{effective} = \frac{\ell n(2)}{T_{1/2}}$$

When a radio-active element has probability to more than one decay then its effective decay constant can be given as

$$\boxed{\lambda_{Effective} = \lambda_1 + \lambda_2}$$

$\Rightarrow$ If probability of disintegration of X to Y or Z are p_1 and p_2

Then we can use

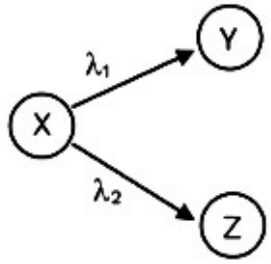
$$\boxed{\lambda_1 = p_1 \times \lambda_{Effective}}$$

$$\boxed{\lambda_2 = p_2 \times \lambda_{effective}}$$

Probability $p_1 = \lambda_1 dt$

$p_2 = \lambda_2 dt$

$p_{Net} = p_1 + p_2 = (\lambda_1 + \lambda_2)dt$



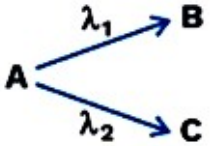
If at t = 0, there are N_0 nuclei of x then after time t number of left nuclei is

$$\boxed{N = N_0 e^{-(\lambda_1 + \lambda_2)\, t}}$$

Rate of production of nuclei of y is $\Rightarrow r_y = \lambda_1 N$

Rate of production of nuclei of z is $\Rightarrow r_z = \lambda_2 N$

Q. **A radioactive nucleus decays into two daughter nuclei with decay constants λ_1 and λ_2 as shown:**



If at t = 0, $N_A = N_0$ and $N_B = N_C = 0$, then find N_B and N_C at any time 't'.

Sol. At t = 0, $N_A = N_0$, $N_B = N_C = 0$
At t = t,

Rate of formation of B, $\frac{dN_B}{dt} = \lambda_1 N_A$ ...(1)

Rate of formation of C, $\frac{dN_C}{dt} = \lambda_2 N_A$ ...(2)

From (1) and (2)

$$\frac{dN_B}{dN_C} = \frac{\lambda_1}{\lambda_2} \quad \Rightarrow \frac{N_B}{N_C} = \frac{\lambda_1}{\lambda_2} \qquad ...(3)$$

Also total number of nuclei will remain constant

$$\therefore \quad N_0 = N_A + N_B + N_C \qquad ...(4)$$

From (3) and (4)

$$N_B = \frac{\lambda_1}{\lambda_1 + \lambda_2}(N_0 - N_A) \text{ and } N_C = \frac{\lambda_2}{\lambda_1 + \lambda_2}(N_0 - N_A) \qquad ...(5)$$

Rate of decay of A, $-\frac{dN_A}{dt} = \lambda_1 N_A + \lambda_2 N_A = (\lambda_1 + \lambda_2)N_A$

On solving,

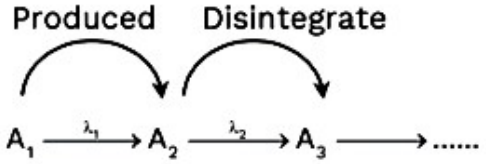
$$N_A = N_0 e^{-(\lambda_1+\lambda_2)t}$$

Putting value of N_A in (5) we get

$$N_B = \frac{\lambda_1}{\lambda_1 + \lambda_2} N_0 (1 - e^{-(\lambda_1+\lambda_2)t})$$

$$N_C = \frac{\lambda_2}{\lambda_1 + \lambda_2} N_0 (1 - e^{-(\lambda_1+\lambda_2)t})$$

SUCCESSIVE DISINTEGRATION OF A RADIOACTIVE ELEMENT IN RADIOACTIVE SERIES

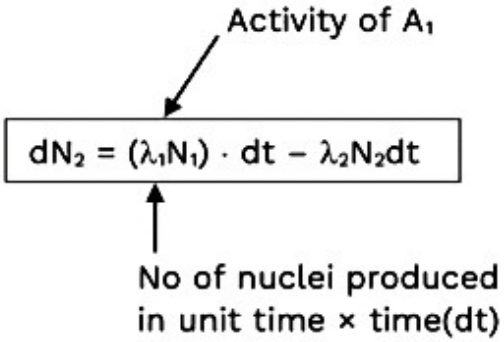


$A_1 \xrightarrow{\lambda_1} A_2 \xrightarrow{\lambda_2} A_3 \longrightarrow \ldots$

Let at t = 0	N_0	0	0
t = t	N_1	N_2	N_3

$N_1 = N_0 \cdot e^{-\lambda t}$

At time t if dN_2 nuclei of A_2 are accumulated in time dt.



So; $dN_2 = (\lambda_1 N_1 - \lambda_2 N_2)$

$$\boxed{\frac{dN_2}{dt} = \lambda_1 N_1 - \lambda_2 N_2}$$

Rate of accumulation of A_2.

$$\frac{dN_2}{dt} + \lambda_2 N_2 = \lambda_1 N_1$$

$$\frac{dN_2}{dt} + \lambda_2 N_2 = \lambda_1 \times \left(N_0 \cdot e^{-\lambda t}\right)$$

It is a linear differential equation in the form of

$$\boxed{\frac{dy}{dx} + Py = Q}$$

After solving this equation we get

$$\boxed{N_2 = \frac{\lambda_1 N_0}{(\lambda_1 - \lambda_2)}\left[e^{-\lambda_2 t} - e^{-\lambda t}\right]}$$

Q. **A factory produces a radioactive substance A at a constant rate R which decays with a decay constant λ to from a stable substance. Find (i) the no. of nuclei of A and (ii) Number of nuclei of B, at any time t assuming the production of A starts at t = 0. (iii) Also find out the maximum number of nuclei of 'A' present at any time during its formation.**

Sol. (i) $\text{Factory} \xrightarrow[\text{const. rate}]{R} A \xrightarrow[\text{decay}]{\lambda} B$

Let N be the number of nuclei of A at any time t

$$\therefore \frac{dN}{dt} = R - \lambda N$$

$$\Rightarrow \int_0^N \frac{dN}{R - \lambda N} = \int_0^t dt$$

On solving we will get $N = R/\lambda\ (1 - e^{-\lambda t})$

(ii) Number of nuclei of B at any time t,

$N_B = Rt - N_A = Rt - R/\lambda(1 - e^{-\lambda t}) = R/\lambda(\lambda t - 1 + e^{-\lambda t})$.

(iii) Maximum number of nuclei of 'A' present at any time during its formation = R/λ.

NUCLEAR REACTIONS

When two nuclei come close to each other, nuclear reaction can occur that results in new nuclei being formed. Generally, it is very difficult to bring nuclei very close as they are positively charged and repulsion between them keeps them beyond the range where they can interact unless they are moving very fast toward each other to decrease the distance of their closest approach to start the nuclear reaction.

In Sun, nuclear reactions are very frequent as the temperature is millions of kelvin, the nuclei have sufficient speed to come very close to each other and the energy released in reaction maintains the temperature there.

A nuclear reaction can occur only if certain conservation laws are followed. These are :

1. Conservation of mass number A.
2. Conservation of charge.
3. Conservation of energy, linear momentum, and angular momentum.

Q. **Find the Q value of the reaction $p + {}^7Li \rightarrow {}^4He + {}^4He$. Find whether the reaction is endothermic exothermic or. The masses of 1H, ^{4}He, and ^{7}Li are 1.007825 u, 4.002603 u, and 7.016004 u, respectively.**

Sol. Given is the total mass of the initial particles, $m_i = 1.007825 + 7.016004 = 8.023829$ u

and the total mass of the final particles, $m_f = 2 \times 4.002603 = 8.005206$ u

The Initial and final mass of particles have a difference of

$\Delta m = m_i - m_f = 8.023829 - 8.0065206 = 0.018623$ u

This mass is converted into energy and the reaction is exothermic.

The Q value is positive and given by, $Q = (\Delta m)c^2 = 0.018623 \times 931.5 = 17.35$ meV

Q. **Show that $^{230}_{92}U$ does not decay by emitting a neutron or proton. Given :**

$M(^{229}_{92}U) = 230.033927\,amu$, $M(^{230}_{92}U) = 229.033496\,amu$,

$M(^{229}_{91}Pa) = 229.032089\,amu$, $m(n) = 1.008665\,amu$,

$m(p) = 1.007825\,amu$

Sol. The corresponding decay equations would be

(a) $^{230}_{92}U \rightarrow n + ^{229}_{92}U$ (b) $^{230}_{92}U \rightarrow p + ^{229}_{91}Pa$

In the first equation, the energy released can be given by

$$Q = [230.033927 - 229.033496 - 1.008665] \times 931.5 = -7.7\ MeV$$

Because $Q < 0$, neutron decay is not possible spontaneously.

Similarly, in the second equation, the energy released can be given as

$$Q = [230.033927 - 229.032089 - 1.007825] \times 931.5 = -5.6\ MeV$$

As $Q < 0$, proton decay is also not possible spontaneously

Q. **Estimate the minimum kinetic energy of an α-particle to cause the reaction $^{14}N(\alpha, p)\ ^{17}O$. Given masses of ^{14}N, ^{4}He, ^{1}H and ^{17}O are 4.00260 u, 1.00783 u and 16.99913 u.**

Sol. $^{14}N + {}^{4}He \longrightarrow {}^{1}H + {}^{17}O$

Thus, we have

$$Q = (14.00307\,u + 4.00260\,u - 1.00783\,u - 16.9913\,u)\left(931.5\,\frac{MeV}{u}\right)$$

$$= -1.20\ MeV$$

Q value is negative. It means reaction is endothermic. So, the minimum kinetic energy of α-particle to initiate this reaction would be

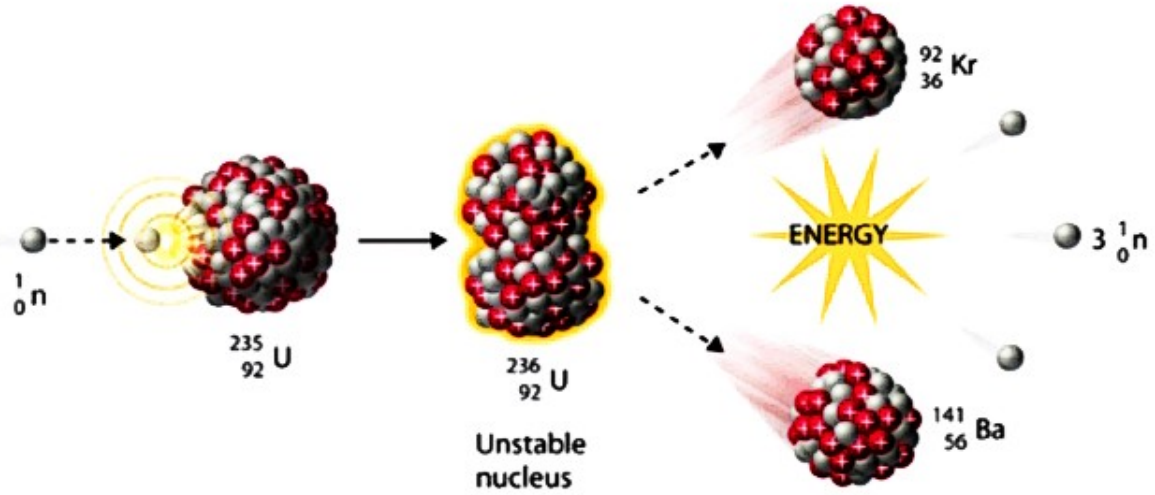
$$K_{min} = |Q|\left(\frac{m_\alpha}{m_N} + 1\right) = (1.20)\left(\frac{4.00260}{14.00307} + 1\right) = 1.54\ MeV$$

There are two major categories of nuclear reaction, we will discuss in detail. There are

(i) Nuclear fission (ii) Nuclear fusion

Nuclear Fission

Nuclear fission is a reaction in which the nucleus of an atom split into two or more smaller nuclei by bombarding nuclei with other nuclear particle such as proton, neutron or α-particle etc.



When a uranium isotopes $^{235}_{92}U$ bombarded with a neutron breaks into two intermediate mass nuclear fragments.

$$_0n^1 + {}^{235}_{92}U \longrightarrow {}^{236}_{92}U \longrightarrow {}^{141}_{56}Ba + {}^{92}_{36}Kr + 3\,{}^{1}_{0}n$$

$$Q\,\text{value} = \left[(MU - 92m_e + m_n) - \{(M_{Ba} - 56M_e) + (M_{Kr} - 36M_e + 3M_n)\}\right]C^2$$

Q value = $(\Delta m)C^2$.

Important Points :-

The same reaction can produce other pair of intermediate mass fragments.

(1) $_0n^1 + {}^{235}_{92}U \longrightarrow {}^{236}_{92}U \longrightarrow {}^{144}_{56}Ba + {}^{89}_{36}Kr + 3\,{}^{1}_{0}n$

(2) $_0n^1 + {}^{235}_{92}U \longrightarrow {}^{236}_{92}U \longrightarrow {}^{133}_{51}Sb + {}^{99}_{41}Nb + 4\,{}^{1}_{0}n$

(3) $_0n^1 + {}^{235}_{92}U \longrightarrow {}^{140}_{54}Xe + {}^{94}_{38}Sr + 2_0n^1$

The fragment produced are radio-active nuclei ; they emit β-particles in succession to active stable end product.

Chain reaction is a very important and interesting feature of neutron induced fission.

The energy released in the fission of uranium of the order of 200 MeV.

Q. **Calculate the total energy released if 1.0 kg of ^{235}U undergoes fission, taking the disintegration energy per event to be Q = 208 MeV (a more accurate value than the estimate given previously).**

Sol. We need to know the number of nuclei in 1.00 kg of uranium. Because A = 235, the number of nuclei is

$$N = \left(\frac{6.02 \times 10^{23} \text{ nuclei mol}^{-1}}{235 \text{ g mol}^{-1}} \right) (1.00 \times 10^{3} \text{ g})$$

$$= 2.56 \times 10^{24} \text{ nuclei}$$

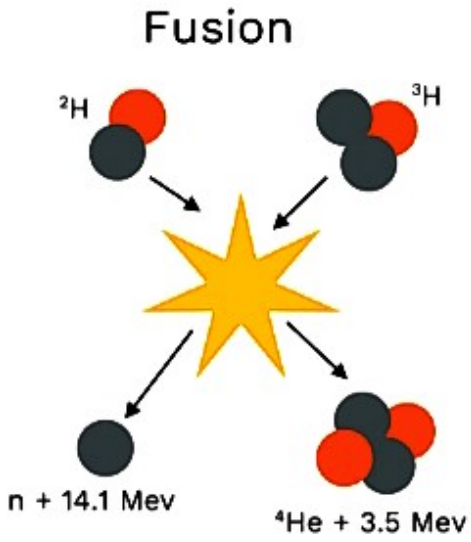
Hence, the disintegration energy is

$$E = NQ = (2.56 \times 10^{24} \text{ nuclei}) \left(208 \frac{\text{MeV}}{\text{nucleus}} \right)$$

$$= 5.32 \times 10^{26} \text{ MeV}$$

Nuclear Fusion :- (Energy Generation in Stars)

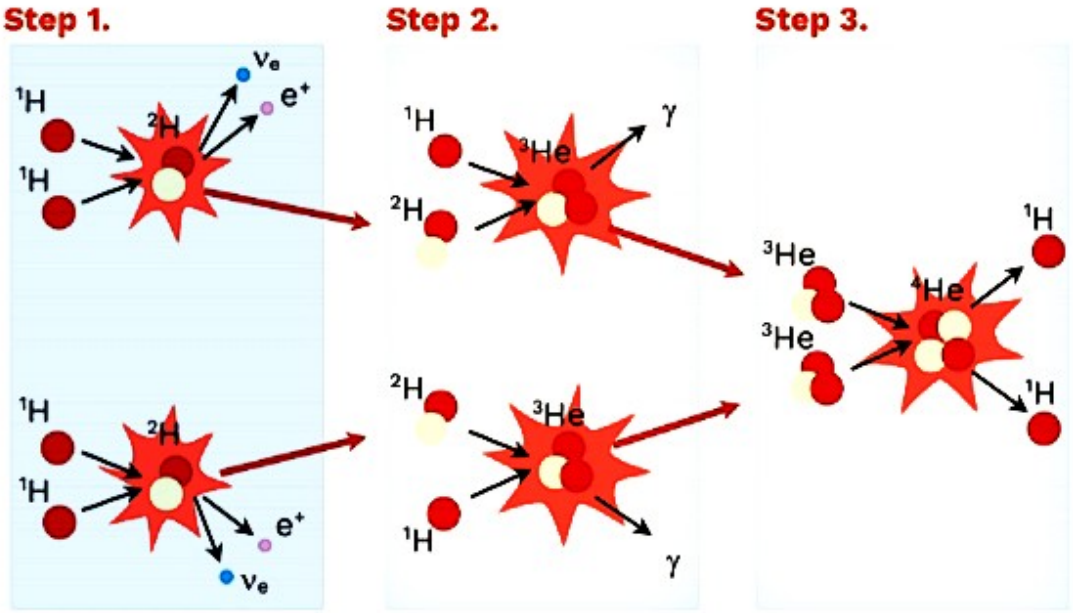
Nuclear fusion is a reaction in which two or more atomic nuclei are combined to form one or more different atomic nuclei and subatomic particle.



For fusion to take place the two nuclei must come close enough so that attractive short range nuclear force is able to affect then.

However, since they are both positively charge particle, they experience coulomb repulsion. They therefore, must have enough energy to overcome this coulomb barrier. The temperature required to cross this barrier is approximately 3×10^9 K.
When fusion is achieved by raising the temperature of the system so that particles have enough kinetic energy to overcome the coulomb repulsive behaviour, called **Thermo Nuclear Fusion**. Thermonuclear fusion is the source of energy output in the interior of stars.

Fusion Reaction of Sun :-
In fusion reaction of Sun, Hydrogen burned into Helium. Hydrogen is the fuel of sun.



Step 1. Hydrogen + Hydrogen ⟶ Deuterium + Positron + Neutrino

Step 2. Deuterium + Hydrogen ⟶ Helium-3 + Photon

Step 3. Helium-3 + Helium-3 ⟶ Helium-4 + Hydrogen + Hydrogen

There are following reactions takes place in the core of the Sun.

$$_1^1H + {}_1^1H \rightarrow {}_1^2H + e^+ + v + 0.42\,MeV \quad ...(i)$$

$$e^+ + e^- \rightarrow \gamma + \gamma + 1.02\,MeV \quad ...(ii)$$

$$_1^2H + {}_1^1H \rightarrow {}_2^3He + \gamma + 5.49\,MeV \quad ...(iii)$$

$$_2^3He + {}_2^3He \rightarrow {}_2^4He + {}_1^1H + {}_1^1H + 12.86\,MeV \quad ...(iv)$$

Thus, four Hydrogen atoms combine to form an $_2He^4$ atom with a release of 26.7 MeV of energy.

Q. **A deuterium reaction that occurs in an experimental fusion reactor is in two stages :**

(i) Two deuterium ($_1^2D$) nuclei fuse together to form a tritium nucleus, with a proton as a by product written as D(D, p)T.

(ii) A tritium nucleus fuses with another deuterium nucleus to form a helium $_2^4He$ nucleus with neutron as a by-product, written as T(D, n) $_2^4He$.

Compute :

(a) the energy released in each of the two stages,

(b) the energy released in the combined reaction per deuterium.

(c) What percentage of the rest mass energy of the initial deuterium is released in the reaction. Given,

$_1^2D = 2.014102\,amu$; $_1^3T = 3.016049$; $_2^4He = 4.002603\,amu$;

$_1^1H = 1.007825\,amu$; $_0^1n = 1.00665\,amu$

Sol. (a) (i) The reaction involved in the process is

$$_1^2D + {}_1^2D \rightarrow {}_1^3T + {}_1^1p + E_1$$

Mass defect of the above reaction is

$$\Delta m = [2(2.014102) - (3.016049 + 1.007825)] = 0.00433\,amu$$

Energy released in the process is

$$E_1 = 0.00433 \times 931.5\ MeV = 4.033\ MeV$$

(ii) The reaction involved in the process is

$$_1^3T + {}_1^2D \rightarrow {}_2^4He + {}_0^1n + E_2$$

Mass defect of above reaction is

$$\Delta m = [(3.016049 + 2.014102) - (4.002603 + 1.008665)] = 0.01888\ amu$$

Energy released in the process is

$$E_2 = 0.01888 \times 931.5\ MeV = 17.587\ MeV$$

Total energy released in both processes is

$$17.587 + 4.033 = 21.62\ MEV$$

(b) $\because$ 3 deuterium atoms are involved in energy calculation.

$\therefore$ Energy released per deuterium atom is

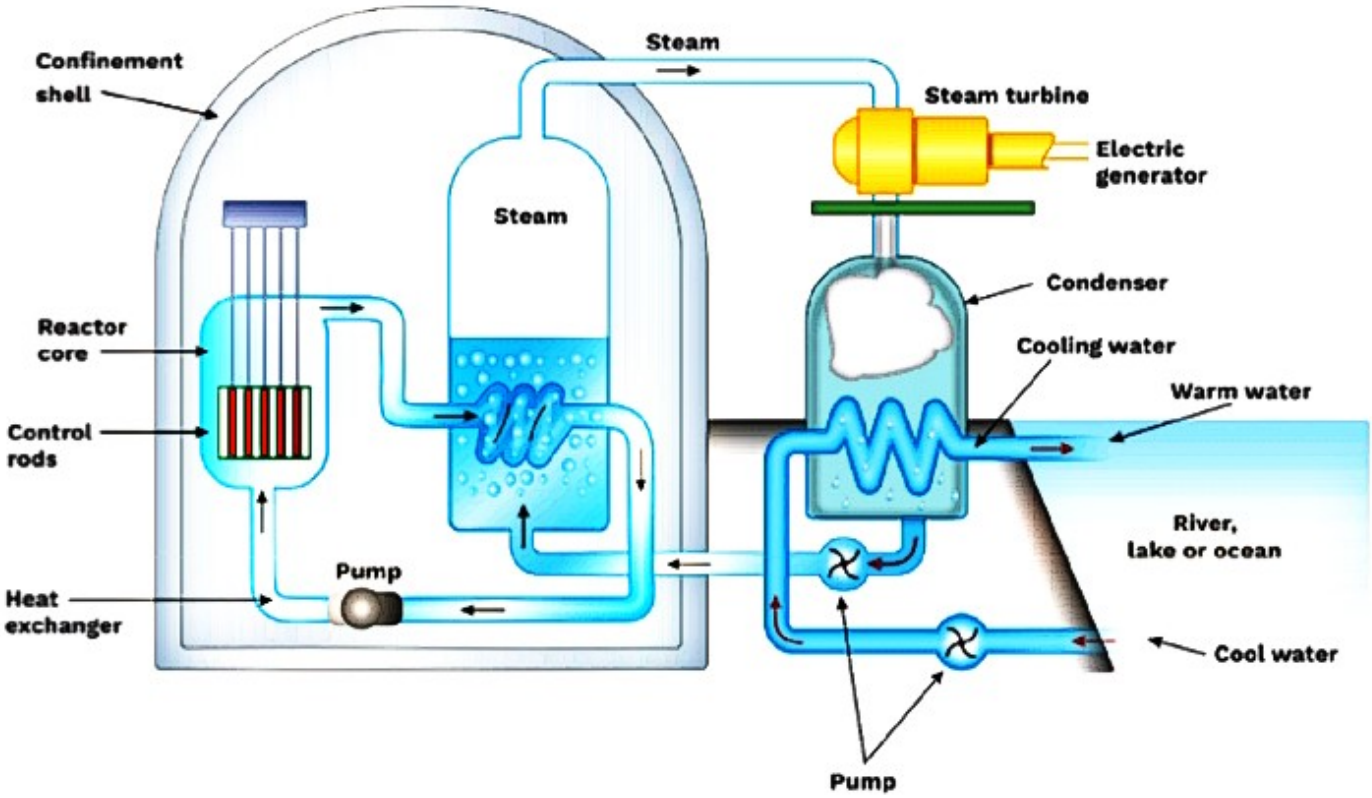
$$E' = \frac{21.62}{3} = 7.207\ MeV$$

(c) % of rest mass energy of $_1^2D$ released in the reaction is

$$f = \frac{7.207}{2.014102 \times 931.5} = 0.384\%$$

NUCLEAR REACTOR

A Nuclear reactor is a device used to initiate and control a fusion nuclear chain reaction or nuclear fission chain reaction. The heat generate during reaction used to runs steam turbine to generate electricity also used in nuclear submarine propulsion.



Important components of a nuclear reactor:

(i) Moderators: The average energy of the neutrons released during the fission of a 92U235is 2 Mev. These neutrons, unless they are slowed down, will exit the reactor without interacting with the uranium nuclei. Fast neutrons must be slowed down so they can be absorbed by uranium

When a neutron is made to strike a light nucleus like the nucleus of hydrogen, it loses almost everything of its kinetic energy.

(ii) Multiplication factor (K): The ratio of number of fissions produced by given generation of neutrons to the number of fissions of the preceding generation is called multiplication factor.

If K = 1, the operation of reactor is said to be critical.

For steady generation of power K must be equal to 1

If K > 1, the reaction rate and reactor power increases exponentially and if K is not brought down the reactor will become super critical and may explode.

(iii) Control Rods: The reaction rate is controlled through control-rods made out of neutron absorbing material such as cadmium.

(iv) Safety Rods: These rods are provided in reactors in addition to control rods. These rods, when required, can be inserted into the reactor and K can be reduced.

Q. **A 1000 MW fission reactor consumes half of its fuel in 5.00 yrs. How much did it contain initially? Assume that the reactor operates 80% of the time, that all the energy generated arises from the fission of $^{235}_{92}U$ and this nuclei is consumed only by the fission process.**

Sol. In the fission of one nucleus of 1 kg of $^{235}_{92}U$, energy generated is 200 MeV.

Energy generated in the fission of 1 kg of $^{235}_{92}U$

$$= (6.023 \times 10^{23}) \left(\frac{1000\ g}{235\ g} \right) (200 \times 1.6 \times 10^{-13} J)$$

$$= 8.2 \times 10^{23}\ J/kg$$

Time for which the reactor actually operates

$$= \left(\frac{80}{100} \right) \times 5\ yr = 4\ yr = 4 \times (3.154 \times 10^{7}\ s) = 12.62 \times 10^{7}\ s$$

Energy generated by the reactor in 5y = power × time

$$= (1000 \times 10^{6}\ W)\ (12.62 \times 10^{7}\ s) = 12.62 \times 10^{16}\ J$$

Fuel (U^{235}) consumed in 5 years $= \dfrac{12.62 \times 10^{16} J}{8.2 \times 10^{13} J/kg} = 1.5 \times 10^{3}\ kg$